SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE: ARTIFICIAL INTELLIGENCE **COURSE CODE: CSA1703**

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning,

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Hashini S / 192524035 (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Hashini S

ARTICLE REVIEW REPORT

Selected Article

Badr, M. M., Hamad, M. S., Abdel-Khalik, A. S., Hamdy, R. A., Ahmed, S., & Hamdan, E. (2021). Fault Identification of Photovoltaic Array Based on Machine Learning Classifiers. IEEE Access, 9, 159113–159132. DOI: 10.1109/ACCESS.2021.3130889.

The selected article is a peer-reviewed IEEE Access research paper published in 2021, satisfying the assessment requirement of a post-2019 research article. The paper studies supervised machine-learning techniques for automatic fault detection and diagnosis in grid-connected photovoltaic (PV) arrays. The authors compare Decision Tree (DT), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) classifiers, with Bayesian Optimization used for hyperparameter selection. The work contains both simulation-based and experimental evaluation.

Abstract of the Review

This review examines the motivation, methodology, datasets, machine-learning models, evaluation strategy, findings, limitations, and real-world relevance of the selected research article. The article addresses a practical classification problem: identifying whether a PV array is healthy or faulty and, after detection, determining the type of fault. The study is closely connected to CO4 because it demonstrates decision-tree learning and supervised classification for decision-making in an engineering environment. It also provides a useful comparison with statistical and distance-based classifiers. The review emphasizes that very high accuracy obtained on simulated data does not automatically guarantee equivalent performance in real deployment. Experimental validation is therefore important. The article's use of multiple environmental and electrical operating conditions, Bayesian hyperparameter optimization, k-fold cross-validation, hold-out testing, and a separate experimental dataset strengthens the methodology. At the same time, the dependence on a controlled PV configuration and simulated scenarios creates a possible generalization limitation. A future extension should evaluate the models on larger, geographically diverse, naturally occurring PV-fault datasets and examine robustness under unseen operating conditions.

Keywords

- Artificial Intelligence
- Decision Tree
- Machine Learning
- Photovoltaic Fault Detection
- Fault Diagnosis
- KNN
- SVM
- Bayesian Optimization
- Classification
- CO4

TASK 1 – ARTICLE SUMMARY

1(a) Full Citation, Domain and ML Problem

Full citation: Mohamed M. Badr, Mostafa S. Hamad, Ayman S. Abdel-Khalik, Ragi A. Hamdy, Shehab Ahmed, and Eman Hamdan, "Fault Identification of Photovoltaic Array Based on Machine Learning Classifiers," IEEE Access, Volume 9, pp. 159113–159132, 2021. DOI: 10.1109/ACCESS.2021.3130889.

Domain/Application Area: Renewable energy systems, photovoltaic power generation, electrical fault monitoring, and intelligent condition monitoring.

Specific ML problem: The paper develops a supervised machine-learning fault detection and diagnosis (FDD) framework for a grid-connected PV array. Detection separates normal/healthy operating conditions from faulty conditions. Diagnosis then distinguishes among fault categories such as arc fault, MPPT unit failure, line-to-line fault, open-circuit fault, and partial shading. The problem is therefore a multi-stage classification and decision-making task.

Research Context

PV systems are increasingly deployed in electrical grids. Although PV arrays generally have low maintenance requirements, electrical faults can reduce energy production, damage equipment, and create safety concerns. Conventional protection devices such as overcurrent protection and arc-fault protection are necessary, but they may not provide sufficient diagnostic information to identify the exact fault type. A monitoring system that can interpret electrical and environmental measurements can reduce troubleshooting time and support faster maintenance decisions.

1(b) Article Objective and Research Gap

The central objective of the article is to develop an automatic machine-learning-based system that can detect and diagnose important PV-array faults using a small set of measurable variables. The authors focus on four input quantities: solar irradiance, temperature, PV-array voltage, and PV-array current. These variables are available from a PV monitoring arrangement and can be used to characterize the operating state of the array.

The research gap arises from limitations in conventional fault protection and from the incomplete coverage of earlier machine-learning studies. Traditional protection can indicate abnormal electrical behavior but may not distinguish different fault categories reliably, especially when irradiance, temperature, fault impedance, and mismatch conditions vary. Earlier approaches may also require more sensors or focus on a smaller set of fault scenarios. The authors address this gap by comparing multiple supervised classifiers under a common FDD framework, optimizing their hyperparameters using Bayesian Optimization, considering several fault conditions, and validating the selected models experimentally.

In my interpretation, the major contribution is not simply choosing one classifier. Instead, the study creates a structured decision pipeline in which detection and diagnosis are treated as separate classification tasks. This separation makes it possible to select a model appropriate to each task and provides a practical path from data acquisition to an operational fault decision.

TASK 2 – METHODOLOGY ANALYSIS

2(a) Machine-Learning Techniques

The article evaluates three supervised machine-learning families: Decision Tree, KNN, and SVM. The classifiers are tested with different internal configurations so that the researchers can identify suitable models for both fault detection and fault diagnosis.

Decision Tree

The Decision Tree models are based on the CART approach and use three splitting criteria: Gini Index, Towing Rule, and Deviance. A decision tree repeatedly divides the feature space according to feature thresholds. Each internal node represents a decision, and each leaf represents a predicted class. This is directly related to CO4 because decision trees are inductive learning models: the algorithm learns decision rules from labeled examples and uses those rules to classify unseen observations.

A major advantage of the tree is interpretability. For example, a learned rule can conceptually resemble: if a measured PV operating variable is below a threshold, follow one branch; otherwise follow another branch. Such rules can be easier for engineers to inspect than a complex black-box model.

K-Nearest Neighbors

KNN classifies a new sample according to the labels of nearby training observations. The study considers Euclidean, City Block, Mahalanobis, and Cosine distance metrics. Bayesian Optimization is used to select suitable parameters. KNN requires standardized data because distance calculations are sensitive to the scale of input variables.

Support Vector Machine

SVM seeks a decision boundary that separates classes while maximizing the margin between them. The article evaluates Linear, Quadratic, Cubic, and Gaussian Radial Basis (GRB) kernels and considers different multiclass approaches for diagnosis. SVM can be effective when classes are not linearly separable and a suitable kernel transformation is selected.

Bayesian Optimization

Bayesian Optimization is used for hyperparameter tuning. Instead of manually testing every possible combination, it uses information from previously evaluated configurations to guide the search toward promising configurations. This reduces the need for exhaustive trial-and-error and helps identify high-performing models.

TASK 2 – DATASET AND PREPROCESSING

Dataset Description

The study uses a 4 kW grid-connected PV array represented by four parallel PV strings, with four series PV modules in each string. The simulated dataset contains 14,100 samples. The study varies solar irradiance, temperature, shading, and fault impedance to create a broad set of operating conditions.

Condition	Fault/Scenario	Samples	Irradiance	Temperature
Healthy	Normal condition	564	50–1200 W/m²	0–55°C
Fault	Arc fault	2820	50–1200 W/m²	0–55°C
Fault	MPPT unit failure	2820	50–1200 W/m²	0–55°C
Fault	Line-to-line	2820	50–1200 W/m²	0–55°C
Fault	Open circuit	2820	50–1200 W/m²	0–55°C
Fault	Partial shading	2256	50–1200 W/m²	0–55°C

The four primary input quantities are solar irradiance G, temperature T, PV-array voltage V_PV, and PV-array current I_PV. The authors also calculate PV output power and a derived quantity gamma, defined as PV output power divided by solar irradiance. The model therefore uses physically meaningful measurements rather than arbitrary numerical features.

Data Preparation

- Raw measurements are converted into a meaningful structured format.
- Missing values and records outside the required time threshold are handled.
- Categorical case labels are represented numerically.
- Standardization is applied to KNN and SVM because their calculations depend on feature scale.
- Decision Trees do not require feature rescaling because tree splits compare values against thresholds.
- The dataset is divided using a 65% tuning/training portion and a hold-out portion for generalization testing.
- K-fold cross-validation is used during model selection and Bayesian hyperparameter optimization.

Experimental Dataset

In addition to simulation, the authors conduct an experimental study. The experimental system collects 1,362 fault-related observations, with 227 instances for each of six classes: free-fault, arc fault, line-to-line fault, open-circuit fault, partial shading, and MPPT unit failure. Measurements include PV voltage, current, solar irradiance, and temperature. The experimental data are important because they test whether the simulation-derived approach can operate under physical conditions.

TASK 2 – CO4 MAPPING AND METHODOLOGICAL EVALUATION

Mapping the Article to CO4

CO4 Topic	Evidence in Article	Concept Demonstrated	Relevance
Inductive Learning	Models learn class boundaries/rules from labeled PV samples.	Generalization from examples	High
Decision Trees	CART trees with Gini, Towing Rule and Deviance.	Recursive partitioning and classification	Very High
Statistical Learning	KNN and SVM provide alternative supervised classifiers.	Distance-based and margin-based classification	High
Classification	Detection and diagnosis are treated as classification tasks.	Multi-stage decision-making	Very High
Hyperparameter Optimization	Bayesian Optimization selects model configurations.	Model selection	High
Model Evaluation	Cross-validation and hold-out testing are used.	Generalization assessment	Very High

Methodological Strength

A major methodological strength is the combination of model comparison, hyperparameter optimization, cross-validation, hold-out testing, and experimental validation. The study does not simply report one accuracy value from one classifier. It compares several configurations and separates detection from diagnosis. This is valuable because detecting "fault versus no fault" is generally easier than identifying the exact fault category. The separate stages reflect the actual engineering decision process.

Methodological Limitation

The main limitation is the dependence on a controlled PV configuration and generated simulation data. Although the experimental dataset strengthens the work, the simulation dominates the overall dataset size. Real PV installations vary in module technology, inverter behavior, sensor noise, weather patterns, installation geometry, aging, and fault combinations. Therefore, a model that performs extremely well on a controlled simulated environment may not maintain the same performance on unseen installations. The diagnosis stage also shows a noticeable performance reduction compared with detection, indicating that distinguishing among fault types is substantially harder.

Why the Limitation Matters

A model can learn patterns that are strongly tied to the simulator or a particular PV configuration. If the operating distribution changes, the learned decision boundaries may no longer be optimal. This is a classic generalization problem in machine learning. A strong future study should use cross-site and cross-device validation, where a model is trained on one set of PV systems and tested on another.

TASK 3 – RESULTS AND COMPARATIVE EVALUATION

3(a) Key Results

The article reports very high detection performance for several models. In the simulation test results, Decision Tree detection models achieved approximately 99.73–99.79% test accuracy. KNN detection models achieved approximately 99.97–100% for the strongest configurations. SVM detection models achieved 99.97–100% test accuracy. The diagnosis task was more difficult: Decision Tree diagnosis reached about 85.68% for its strongest tested configurations, while KNN diagnosis was lower, around 81.73% for the strongest SVM-style comparison context and around the high-70% range for KNN configurations. The experimental results provide a more realistic check.

Model family	Best/representative detection test accuracy	Best/representative diagnosis test accuracy	Observation
Decision Tree	99.79%	85.68%	Highly accurate detection; diagnosis is harder.
KNN	≈100%	≈80% range	Excellent detection but lower diagnosis performance.
SVM	100%	≈89.84% in reported model set	Strong overall classifier; kernel choice matters.

For the physical experimental dataset, the paper reports detection accuracies of 98.31% for the Gini Decision Tree, 98.31% for the Towing Rule Decision Tree, 99.15% for the Deviance Decision Tree, 95.58% for Euclidean KNN, 96.00% for City Block and Mahalanobis KNN, 92.43% for Cosine KNN, and 99.36% for the Linear SVM detection model. The experimental diagnosis values are lower, including 83.37%, 85.89%, and 86.14% for the three Decision Tree criteria, and 85.64% for the Linear SVM model. These results show that performance decreases when the system moves from controlled simulation to physical measurements.

Detection versus Diagnosis

The distinction between detection and diagnosis is one of the most important findings. Detection answers: “Is the PV system abnormal?” Diagnosis answers: “Which fault caused the abnormal behavior?” A healthy-versus-fault classifier can exploit broad differences in electrical behavior. Diagnosis must discriminate among faults whose electrical signatures may overlap, particularly under different irradiance, temperature, shading, and impedance conditions.

TASK 3 – CRITICAL EVALUATION

3(b) Are the Reported Metrics Realistic?

The reported metrics are plausible within the experimental design, but they should not be interpreted as universal PV fault-classification accuracy. Detection accuracies near 100% are possible when the simulation produces clearly structured class patterns and when the test data are drawn from the same overall distribution as the training data. The experimental results are slightly lower and therefore provide a useful reality check. In particular, the reduction in diagnosis accuracy indicates that the fault classes overlap more strongly in real measurements.

Potential Dataset Bias

The simulation uses a specific 4 kW PV configuration and a predefined range of irradiance, temperature, shading, and fault impedance. This creates a controlled but limited distribution. Real-world systems may include different module ratings, inverter controls, sensor calibration errors, communication delays, module degradation, dust, snow, rapidly changing cloud cover, and combined faults. The dataset also contains many more simulated observations than experimental observations. Consequently, the headline accuracy is likely to be more representative of the studied configuration than of every PV plant.

Potential Evaluation Bias

Random or distribution-preserving splits can allow training and testing examples to remain highly similar. A stronger test would use scenario-level or site-level separation: all observations from one weather day, PV installation, fault impedance range, or hardware configuration should be held out. This would reveal whether the classifier can generalize to a genuinely unseen environment.

Additional Experiments Recommended

- Leave-one-weather-condition-out validation, such as training on clear and partly cloudy days and testing on unseen weather patterns.
- Cross-site validation using different PV array ratings, module technologies, and inverter manufacturers.
- Noise robustness experiments by adding realistic sensor and communication noise.
- Hybrid-fault experiments where two abnormal conditions occur simultaneously.
- Ablation study to measure the individual contribution of irradiance, temperature, voltage, and current.
- Concept-drift testing to study performance as PV modules age and their electrical characteristics change.

Critical Overall Judgment

Overall, the methodology is technically strong and clearly demonstrates the usefulness of supervised machine learning for PV fault detection. The most convincing part is the combination of simulation and physical experimentation. However, the strongest numerical results should be treated as evidence of performance under the study conditions rather than proof of universal robustness.

TASK 3 – REAL-WORLD APPLICABILITY

3(c) Where Can the Technique Be Deployed?

The proposed FDD framework can be deployed in grid-connected solar farms, commercial rooftop PV systems, residential solar installations, microgrids, and renewable-energy monitoring centers. A monitoring controller can continuously collect voltage, current, irradiance, and temperature measurements and pass them to a classification model. The first model can flag abnormal operation. A second model can identify the probable fault category and support a maintenance decision.

Potential Deployment Workflow

1. Sensors collect PV voltage, current, irradiance, and temperature.
2. The monitoring device performs basic cleaning and feature preparation.
3. The detection classifier determines whether the operating state is healthy or abnormal.
4. If abnormal, the diagnosis classifier predicts the likely fault type.
5. The system generates an alert and stores the event for maintenance analysis.
6. An engineer verifies the prediction before carrying out physical repair or isolation.
7. The confirmed maintenance result can later be used as labeled data for model improvement.

Industry Scaling Challenges

Data availability is the first major challenge. Industrial PV plants may have large amounts of sensor data but relatively few confirmed fault events. Faults are rare, and labeling their exact type often requires expert inspection. This creates an imbalanced learning problem.

Compute and latency are also important. SVM hyperparameter optimization can be computationally expensive, while some KNN configurations require storing training samples. A production system may therefore prefer a compact Decision Tree or a carefully optimized ensemble model at the edge, with heavier analysis performed in a central server.

Cybersecurity and reliability must also be considered. A fault-monitoring system connected to a network should protect sensor data, model parameters, and communication channels. Incorrect fault classification could cause unnecessary maintenance or, in a safety-critical situation, inappropriate system isolation.

Ethical issues in this application are less about personal data and more about reliability, accountability, and safe automation. The model should be treated as a decision-support component rather than an unquestionable authority. Human verification and fail-safe electrical protection should remain available.

Economic Impact

Early fault identification can reduce downtime, improve energy yield, shorten troubleshooting time, and support predictive maintenance. The economic value becomes greater for large solar farms where a single undetected fault can reduce production over a long period. A lightweight model that operates close to the PV system could also reduce communication costs and enable faster alerts.

TASK 4 – REFLECTION AND LEARNING OUTCOME

4(a) Three Key Insights Gained

Insight 1 – Decision Trees Learn Explicit Decision Rules

The article helped me understand that a Decision Tree is an inductive learning method because it learns a set of decision rules from labeled training examples. Instead of memorizing individual observations, the tree partitions the feature space using thresholds. This directly connects to the CO4 topic of Decision Tree algorithms. The comparison of Gini Index, Towing Rule, and Deviance also showed that the splitting criterion can influence the resulting model.

Insight 2 – Detection and Diagnosis Are Different ML Problems

I learned that a classification problem can be divided into multiple stages according to the real decision process. Detecting whether a fault exists is not the same as identifying the exact fault. The article achieved much higher accuracy for detection than diagnosis. This demonstrates that model difficulty depends not only on the algorithm but also on how similar the target classes are.

Insight 3 – Generalization Matters More Than a Single Accuracy Number

The study showed why model evaluation should include cross-validation, hold-out testing, and ideally independent real-world data. A model can obtain extremely high accuracy when the test distribution is similar to the training distribution. Experimental validation revealed lower performance, particularly for diagnosis. This connects to the CO4 concept of building reliable classification models and evaluating their decision-making ability on unseen data.

Connection to Course Concepts

Learning	CO4 Concept	Practical Meaning
Learn rules from examples	Inductive Learning / Decision Trees	Training examples become classification rules.
Separate detection and diagnosis	Classification and Decision-Making	Different ML stages can solve different decisions.
Validate beyond one split	Model Evaluation	Cross-validation and independent tests improve confidence.

Personal Learning Reflection

Before reviewing the paper, it was easy to think of accuracy as the main indicator of a machine-learning model. The article made the evaluation process more meaningful by showing that the same algorithm can perform differently for detection and diagnosis and that simulation results may be stronger than physical results. I also understood why feature selection, preprocessing, hyperparameter tuning, and validation are integral parts of a complete machine-learning solution rather than optional additions.

TASK 4 – PROPOSED EXTENSION AND NOVEL EXPERIMENT

4(b) Proposed Research Extension

Proposed enhancement: Develop a cross-site, noise-aware and drift-aware PV fault diagnosis system using an optimized Decision Tree ensemble combined with an online learning strategy.

Research Idea

The proposed extension would collect real PV data from multiple installations instead of relying mainly on one simulated PV configuration. Each site would contribute normal operating data and confirmed fault events. The model would be trained using data from several sites and tested on a completely unseen site. This would create a much stronger generalization experiment.

Experimental Design

8. Collect voltage, current, irradiance, temperature, inverter status and timestamp data from multiple PV installations.
9. Label confirmed events using maintenance records and engineer verification.
10. Create training, validation and test groups by site rather than by random rows.
11. Add controlled sensor-noise experiments to measure robustness.
12. Compare a single Decision Tree, Random Forest, Gradient Boosted Trees, SVM, and a lightweight online-learning model.
13. Measure accuracy, macro-F1, precision, recall, false-alarm rate, detection latency and memory usage.
14. Evaluate concept drift by testing the model across different seasons and PV aging periods.
15. Use explainability methods such as feature importance and tree-path inspection to help engineers understand predictions.

Why It Is Feasible

The experiment is feasible because the original paper already identifies a compact set of useful measurements: irradiance, temperature, voltage, and current. Modern PV monitoring systems commonly record these signals. The proposed experiment therefore extends the existing framework rather than requiring an entirely new sensing architecture. An initial version could be implemented using publicly available PV datasets and then validated with a small real-world pilot dataset.

Expected Impact

The expected impact is improved generalization to unseen PV systems. Instead of optimizing for maximum accuracy on one controlled dataset, the research would optimize for reliable performance across sites and conditions. This would make the model more suitable for industry deployment. It could also identify when a model is becoming unreliable because of seasonal changes, sensor drift, or module aging.

CONCLUSION AND REFERENCES

Conclusion

The reviewed article presents a practical and well-structured application of supervised machine learning to photovoltaic fault detection and diagnosis. Its strongest contribution is the construction of a two-stage FDD framework that compares Decision Tree, KNN, and SVM classifiers and uses Bayesian Optimization for model selection. The study demonstrates that machine learning can distinguish healthy and faulty PV operating states with very high accuracy under the tested conditions.

From the perspective of CO4, the paper provides a clear example of inductive learning and decision-tree classification. It also demonstrates the importance of comparing alternative statistical learning methods, tuning model parameters, and evaluating generalization. The experimental results are particularly useful because they show that diagnosis is more difficult than detection and that physical measurements can reduce performance compared with simulation.

The main critical point is that high simulated accuracy should not be interpreted as universal real-world accuracy. Broader datasets, independent site validation, realistic noise, hybrid faults, seasonal changes, and long-term monitoring would make the findings more robust. With these improvements, an intelligent PV fault-monitoring system could support faster maintenance, higher energy availability, and safer operation of renewable-energy infrastructure.

References

16. Badr, M. M., Hamad, M. S., Abdel-Khalik, A. S., Hamdy, R. A., Ahmed, S., & Hamdan, E. (2021). Fault Identification of Photovoltaic Array Based on Machine Learning Classifiers. IEEE Access, 9, 159113–159132. DOI: 10.1109/ACCESS.2021.3130889.
17. IEEE Access. Article record and publication information for the selected paper. IEEE Xplore document: 9627668.
18. Matzavela, V., & Alepis, E. (2021). Decision tree learning through a Predictive Model for Student Academic Performance in Intelligent M-Learning environments. Computers and Education: Artificial Intelligence, 2, 100035. DOI: 10.1016/j.caeai.2021.100035. (Additional background on decision-tree learning.)
19. Kherif, O., Benmahamed, Y., Teguar, M., Boubakeur, A., & Ghoneim, S. S. M. (2021). Accuracy Improvement of Power Transformer Faults Diagnostic Using KNN Classifier With Decision Tree Principle. IEEE Access, 9, 81693–81701. DOI: 10.1109/ACCESS.2021.3086135. (Additional comparison of ML-based fault diagnosis.)

Source/DOI Link for the Selected Article

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